

NIST initiated a project to evaluate and select quantum-resistant algorithms. The goal is to establish a set of standardised algorithms that can be widely adopted to secure digital communications in the quantum age.

Some of the quantum-safe proprietary VPN solutions include Palo Alto Network's post quantum IKV VPN support, NordVPN, ProtonVPN, ExpressVPN, etc.

Best practices for quantum-safe VPNs

Adopting best practices while deploying quantum-safe VPNs will help to mitigate challenges, and ensure robust security and efficient performance. First, a thorough assessment of the current infrastructure is essential to identify potential compatibility issues and necessary upgrades. This includes evaluating the existing cryptographic protocols and determining how they can be replaced or supplemented with post-quantum algorithms.

Second, it is crucial to implement a phased migration strategy. This approach minimises disruption and allows for gradual adaptation to new cryptographic standards. During this process, maintaining a hybrid environment where traditional and post-quantum algorithms coexist can provide a smoother transition.

Third, continuous monitoring and performance testing are vital to ensure that the integration of post-quantum cryptographic algorithms does not adversely affect the VPN's speed and reliability. Regular updates and patches should be applied to address any vulnerabilities that may arise as the technology evolves.

Lastly, collaboration with industry peers and adherence to emerging standards are key to achieving interoperability and widespread adoption of quantum-safe solutions. Engaging with organisations like NIST and participating in industry forums can provide valuable insights and contribute to the development of robust standards.

By following these best practices, organisations can effectively deploy quantum-safe VPNs and safeguard their digital communications against future quantum threats.

Open source quantum-safe VPN solutions

In addition to proprietary solutions, the open source community is actively contributing to the development of quantum-safe technologies. Open source implementations of post-quantum cryptographic algorithms are critical for transparency, security, and fostering innovation in the field. Projects like Open Quantum Safe (OQS) aim to integrate post-quantum algorithms into widely used encryption libraries, making it easier for developers to adopt quantum-safe practices. Various open source VPN solutions, such as ExpressVPN's Lightway protocol, Kudelski

Security's WireGuard modifications, and Microsoft's PQCrypto-VPN, are pioneering the integration of these algorithms to ensure future-proof security. While these projects are in varying stages of development and testing, they represent crucial steps towards safeguarding data in the quantum era.

While quantum computing poses a significant threat to traditional VPN security, the development of quantum-safe VPNs offers a promising solution. By adopting post-quantum cryptographic algorithms and adhering to emerging standards, we can ensure that VPNs remain a vital tool for securing our digital communications in the age of quantum computing. The collaborative efforts of industry leaders, standards organisations, and the open source community will be crucial in achieving a secure and resilient quantum-safe future. **END** 🐧

References

- Jon Elliott, Post-Quantum Cryptography: Preparing for the Future of Security, <https://auth0.com/blog/post-quantum-cryptography/>
- National Institute of Standards and Technology (NIST), (2016), Post-Quantum Cryptography Project
- Open Quantum Safe (OQS), Open Source Post-Quantum Cryptography, <https://openquantumsafe.org/>
- Shor, P. W. (1994), Algorithms for Quantum Computation: Discrete Logarithms and Factoring, Proceedings of the 35th Annual Symposium on Foundations of Computer Science
- kelVPN, What are the Benefits of a Post-Quantum Encrypted VPN, <https://kelvpn-network.medium.com/what-are-the-benefits-of-a-post-quantum-encrypted-vpn-a81b6fa45534>
- Post-quantum encryption standards are here - a new era for VPN security begins, TechRadar, <https://www.techradar.com/pro/vpn/post-quantum-encryption-standards-are-here-a-new-era-for-vpn-security-begins>
- Verizon explores how Quantum Safe VPNs could protect today's data from tomorrow's hackers, Verizon, <https://www.verizon.com/about/news/verizon-quantum-safe-vpns-data>
- Microsoft, Post-quantum Crypto and VPNs, <https://www.microsoft.com/en-us/research/project/post-quantum-crypto-vpn/>

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